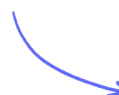


Area & Perimeter

Perimeter / Area / Adding & Subtracting Areas / Problem Solving with Areas

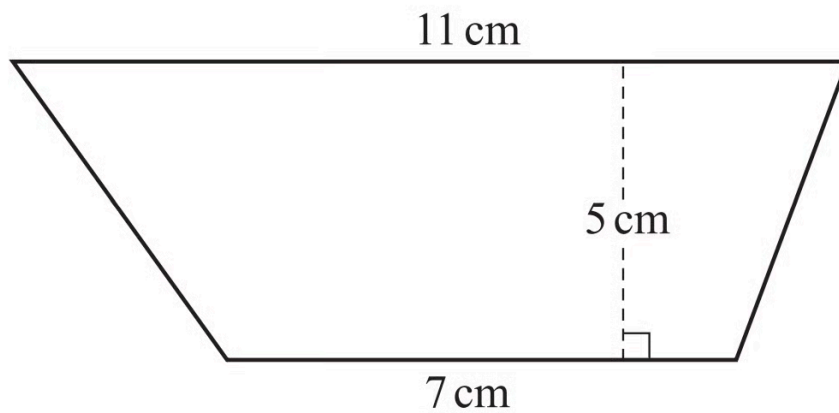
Medium (4 questions)	/7
Hard (4 questions)	/10
Very Hard (4 questions)	/15
Total Marks	/32

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Medium Questions

1



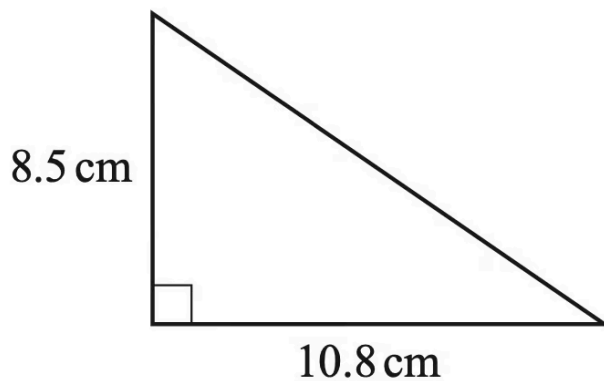
NOT TO
SCALE

Calculate the area of the trapezium.

..... cm²

(2 marks)

2



NOT TO
SCALE

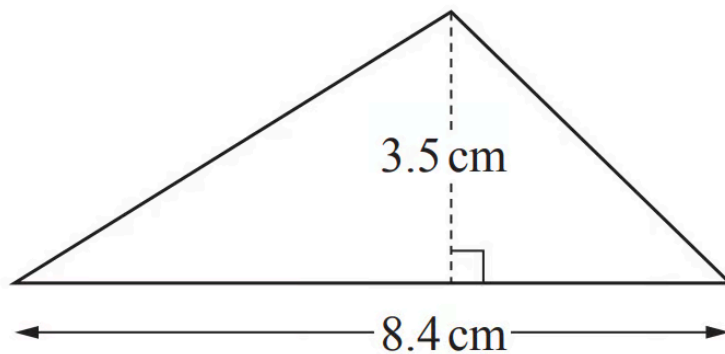
The diagram shows a right-angled triangle.

Calculate the area.

..... cm²

(2 marks)

3



NOT TO
SCALE

Calculate the area of this triangle.

..... cm²

(2 marks)

4 Tom is tiling a floor.

The floor is a rectangle with length 600 cm and width 240 cm
Each tile is a square with side 40 cm

Tom uses this method to work out the number of tiles he needs.

Number of tiles that will fit along the length = $600 \div 40$
= 15
Number of tiles that will fit along the width = $240 \div 40$
= 6
Total number of tiles needed = $15 + 6$
= 21

Give a reason why Tom's method is wrong.

(1 mark)

Hard Questions

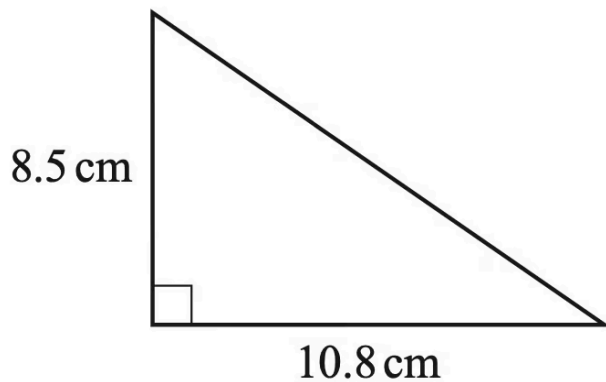
- 1 A rectangle measures 8.5 cm by 10.7 cm, both correct to 1 decimal place.

Calculate the upper bound of the perimeter of the rectangle.

..... cm

(3 marks)

2



NOT TO
SCALE

The diagram shows a right-angled triangle.

Calculate the perimeter.

..... cm

(3 marks)

- 3** The sides of a square are 15.1 cm, correct to 1 decimal place.

Find the upper bound of the area of the square.

..... cm²

(2 marks)

- 4** The area of a square is 42.5 cm², correct to the nearest 0.5 cm².

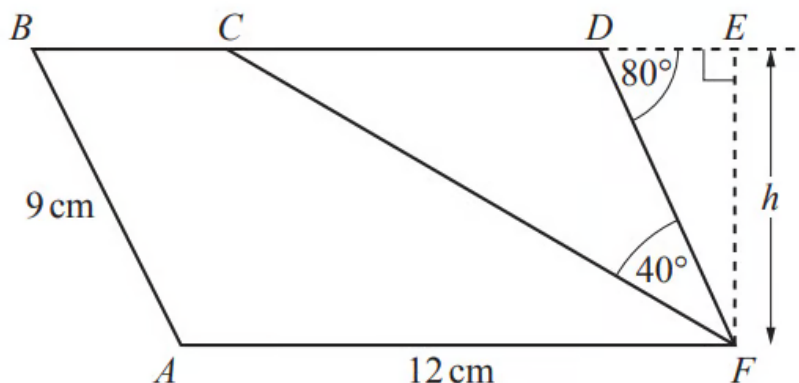
Calculate the lower bound of the length of the side of the square.

..... cm

(2 marks)

Very Hard Questions

1



NOT TO
SCALE

$ABDF$ is a parallelogram and $BCDE$ is a straight line. $AF = 12\text{ cm}$, $AB = 9\text{ cm}$, angle $CFD = 40^\circ$ and angle $FDE = 80^\circ$.

i) Calculate the height, h , of the parallelogram.

$h = \dots\dots\dots\text{ cm}$ [2]

ii) Calculate the area of the **trapezium** $ABCF$.

$\dots\dots\dots\text{ cm}^2$ [3]

(5 marks)

2 The sides of an isosceles triangle are measured correct to the nearest millimetre. One side has a length of 8.2 cm and another has a length of 9.4 cm.

Find the largest possible value of the perimeter of this triangle.

$\dots\dots\dots\text{ cm}$

(3 marks)

- 3 Find the area of a regular hexagon with side length 7.4 cm.

..... cm²

(3 marks)

4

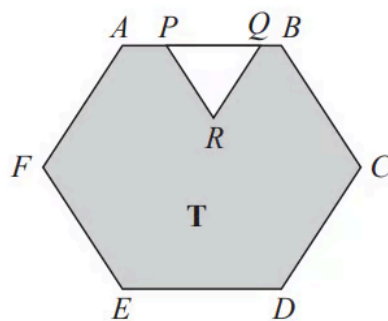


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$.

Given that the area of region **T** is $72\sqrt{3}$ cm² work out the length of PQ .

..... cm

(4 marks)